Worksheet 4.1

Science Quest 7: pages 118–121

Biotic and abiotic factors

Answers

The parts of an ecosystem that are non-living are called the abiotic environment. The parts that are living are called the biotic environment.

1. Complete the table below by classifying the named component as **biotic** or **abiotic**.

Component	Biotic or abiotic?	Component	Biotic or abiotic?
Water salinity	Abiotic	Competitors for mating	Biotic
Wind speed and direction	Abiotic	Producers	Biotic
Competitors for food		Humidity	Abiotic
Predators	Biotic	Decomposers	Biotic
Light penetration	Abiotic	Air temperature	Abiotic

2. The following photo shows a clownfish sheltering amongst the poisonous tentacles of a sea anemone. The sea anemone does not harm the clownfish as it searches for food.

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(a) Identify two abiotic components of this ecosystem.

Water salinity; light penetration

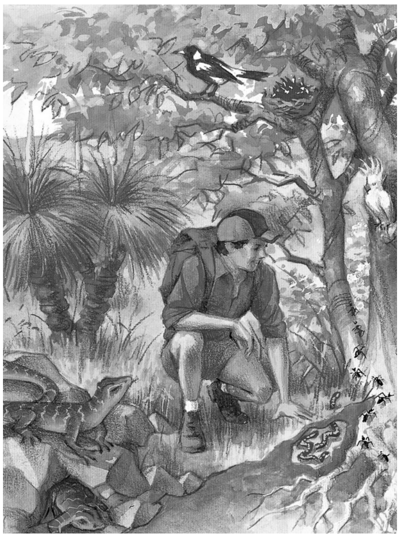
(b) Identify two biotic components of this ecosystem.

Predators; competitors for mating

(c) How does the association of the clownfish with the anemone benefit the fish?

The anemone provides protection from predators.

3. The illustration below shows a boy on a field trip in the Australian bush.



(a) Explain the importance of the trees and grass to this ecosystem.

Trees and grass provide food, habitat and protection.

(b) Earthworms are shown in the soil under the tree. Explain the importance of these worms to the ecosystem.

Earthworms improve the soil quality and help to release nutrients for plants.

(c) The student decides to measure an abiotic factor in this area. Identify an abiotic factor he could measure and explain how he will perform this task.

(A possible response) Humidity could be measured using an hygrometer (wet and dry bulb thermometer). This could be measured in various locations to compare the microclimate of each part of this ecosystem.

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